

CSCC43: Introduction to Databases

Homework - Summer 2024

Due: June 17, 2024

Each exercise is 10 points. Total points 100. Type your answers and submit a PDF file.

Problem 1: Entity-Relationship Model

Construct an E/R diagram for an automobile workshop. The information that the database should contain is the following at minimum:

1. The workshop with name (assume the name identifies the workshop) address and phone number.
2. The automobiles with registration number, make/model, and owner.
3. The customers (automobile owners) with social insurance number, surname, first name, and phone. Each client owns one or more cars.
4. The maintenance work carried out in the workshop with a number (unique in a particular workshop) start and end date of the work, parts replaced (with quantities) and number of labor hours. Spare parts available with code, name and unit cost.

Given the above specification, design the database in the framework of the E/R model. In the design there should be no redundancy of information. The model should consist of the following:

1. Defining the entity sets and the appropriate attributes. You are responsible to define attributes that make sense for this problem. At a minimum you should define and use attributes that lead to a valid E/R design.
2. Defining relationships, and their attributes (if any).
3. Specifying keys for entities and functionality of relationships (many-to-one, etc.).

Extract a relational schema from your design.

Problem 2: Entity-Relationship Model

Give an E/R diagram for the following problem description. Identify entities and relationships and add any attributes to those you judge appropriate. Derive tables from this diagram and for each table identify the primary key. Justify any assumption you might make about the problem.

The application is a hospital ward. We are interested to maintain:

- Each patient with a social insurance number, first and last name and date of birth.
- Admission information including admission date, and doctor in charge of the case. For a discharge the date of the discharge, the discharge reason and for the patient in the ward a contact number for a relative.
- For each doctor their registration number, first and last name, specialization, universities they attended, dates they attended and graduation date.
- Each examination has a date, time, a doctor present, medicines prescribed, and associated dosages, and any diagnoses.
- Each medicine has an identification number, a name and a cost.
- Each illness has a code, a name and a description.

Define the E/R diagram and keys and extract a relational schema. Any assumptions you need to make document them.

Problem 3: Entity-Relationship Model

A University Registrar's office maintains data about the following entities, (a) courses including course number, credits, syllabus, title, and prerequisites, (b) course offerings, including course number, year, semester, section number, instructor(s), timings and classroom, (c) students, including student id, name, program and (d) instructors including id, name, department and title. In addition, the enrollment of students to courses and grades awarded to students for each course they are enrolled for must be appropriately recorded.

Construct an E/R diagram for this organization. List any assumptions you make. Identify the keys and extract the associated relational schema and identify the keys of each relation. In your design, course offerings and courses have to be different entities possibly related.

Problem 4: Functional Dependencies

Use only the Armstrong axioms to show the following:

1. If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$

In the literature this is also known as the Union rule.

2. If $A \rightarrow BC$, then $A \rightarrow B$ and $A \rightarrow C$

In the literature this is also known as the Decomposition rule.

3. If $A \rightarrow B$ and $BC \rightarrow D$, then $AC \rightarrow D$

In the literature this is also known as the Pseudo-transitive rule.

4. Let X be the following set of functional dependencies: $AB \rightarrow CD$, $B \rightarrow DE$, $C \rightarrow F$, $E \rightarrow G$, $A \rightarrow B$

Show that $A \rightarrow FG$ is logically implied by X .

Problem 5: Functional Dependencies

For each of the following schemas and sets of FD's:

1. What are all the nontrivial FD's that follow from the given FD's? You should restrict yourself to FD's with single attributes on the right side.
2. What are all the keys of R ?
3. What are all the superkeys for R that are not keys?

The schemas and dependencies you should consider are:

1. $R(A, B, C, D)$ with FD's $A \rightarrow B$, $B \rightarrow C$, and $B \rightarrow D$.
2. $R(A, B, C, D)$ with FD's $AB \rightarrow C$, $BC \rightarrow D$, $CD \rightarrow A$, and $AD \rightarrow B$.

Problem 6: Functional Dependencies

Consider the following relation and the functional dependencies on it: $R(A, B, C, D, E)$ with $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$

1. Find all candidate keys. Then argue why there cannot be a candidate key with three attributes.
2. Is R in BCNF? Explain.
3. Is R in 3NF?
4. Is the decomposition $R_1(A, B, C)$ and $R_2(A, D, E)$ of R lossless or lossy? Justify your answer. Is this decomposition FD preserving? If your answer is no, what is not preserved? Is the decomposition $R_3(A, B, C, D)$ and $R_4(C, D, E)$ of R lossless or lossy? Justify your answer. Is this decomposition FD preserving? If your answer is no, what is not preserved?
5. For each of R_1 , R_2 , R_3 , R_4 , state if they are in BCNF, 3NF or none.
6. Using the decomposition algorithm, losslessly decompose R into BCNF relations. Then check if it is FD preserving.

Problem 7: Functional Dependencies

Consider a database for investment firm that has attributes:

- ***B*** for Broker
- ***O*** for Office of the Broker
- ***I*** for Investor
- ***S*** for Stock
- ***Q*** for Quantify of stock owned by the investor
- ***D*** for dividend paid by a stock

Thus the overall schema is **$R(\mathbf{B}, \mathbf{O}, \mathbf{I}, \mathbf{S}, \mathbf{Q}, \mathbf{D})$** . Assume that the following functional dependencies are required to hold in the schema: $I \rightarrow B$, $IS \rightarrow Q$, $B \rightarrow O$, $S \rightarrow D$.

- List all the candidate keys for ***R***
- Give a lossless join decomposition of ***R*** into BCNF
- Give a lossless join decomposition of ***R*** into 3NF preserving functional dependencies. Is your final answer in BCNF?

Problem 8: Relational Algebra

Consider the following schema (key underlined):

- *students*(sno, name, addr)
- *universities*(uname, addr)
- *experiment*(eno, num-of-p, date)
- *study*(sno, uname)
- *took-place*(uname, eno)
- *participate*(sno, eno)

The schema records information about *students* who *study* at *universities*. Several *experiments* *took-place* at *universities* on a specific date, on which a number of people *participated*. Based on the above schema, answer the following queries in relational algebra.

- Find the name of a student with student number (*sno*) 5.
- Find the address of the university that the student with the name 'jones' studies.
- Find the universities for which at least one student studying there participated in at least one of the experiments.

- Find the universities for which no students studying there participated in any experiments.
- Find all the experiments on which only students studying at the universities the experiment took place, participated.

Problem 9: Relational Algebra

Consider the following classic schema (keys underlined):

- *sales*(dept, *item*, *volume*)
- *dept*(dname, *floor*)
- *item*(iname, *type*, *color*)
- *employee*(eno, *ename*, *mgr*, *dept*, *salary*, *job-status*)
- *supplier*(sno, *sname*)
- *supply*(supplier, dept, item, *volume*)

Express the following queries in relational algebra:

- Find the items sold by no department on the second floor
- Find the departments where all the employees earn less than their manager.
- Find the items sold by at least two departments on the second floor.
- List the items sold only by departments that sell all the items they are supplied.

Problem 10: Relational Algebra

Consider the database (keys underlined):

- *Representative*(number, *surname*, *firstname*, *committee*, *county*, *constituency*)
- *Consituencies*(county, number, *name*)
- *Counties*(code, *name*, *region*)
- *Committies*(number, *name*, *president*)

Formulate the following queries in relational algebra:

- Find the name and surname of the presidents of the committees in which there is at least one representative from the county of York.
- Find the name and last name of the members of the finance committee.
- Find the name and surname, county and region of the election of the delegates on the finance committee.
- Find the regions in which representatives having the same surname have been elected.